
Algorithm 1: A numbered, ruled, nested algorithm

Data: A list $a_1, \dots, a_n$ and a target x

Result: An index i with $a_i = x$, or $\perp$

1 **Function** *Locate*(a, x):

2 $i \leftarrow 1$;

3 **while** $i \leq n$ **do**

4 **if** $a_i = x$ **then**

5 $\perp$ **return** i ;

6 **else if** $a_i > x$ **then**

7 $\perp$ **break**;

8 $i \leftarrow i + 1$;

9 $\perp$ **return** $\perp$;

Line 2 initializes the invariant; line 8 advances it.